

Hamiltonian Reconstruction as a Physical Theory Object

A typed reconstruction of finite-dimensional Hamiltonian mechanics,
with its composition laws, its empirical interpretation, and its limits

PRINCIPIA PHYSICA pilot, Topic 2

Compositional Foundations of Finite-Dimensional Classical Mechanics

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Abstract

We reconstruct finite-dimensional Hamiltonian mechanics as an explicitly *typed* object — a kinematic type, an observable type, a generator, a dynamical law, and an empirical interpretation — under six stated axioms and one stated assumption. After proving the standard structural facts within this presentation, we use the typing to isolate four results bearing on the compositional thesis under test in this pilot. (i) The Cartesian monoidal product of Hamiltonian theory objects has image exactly the non-interacting composites, so interaction is provably outside the range of $\otimes$. (ii) Symplectic morphisms are immersions, so coarse-graining is not a morphism of the category and reduction must be a different operation. (iii) A globally asymptotically stable dissipative field on a surface is not Hamiltonian for *any* symplectic form, not merely for the standard one. (iv) A matched pair of counterexamples separates mathematical from empirical equivalence in both directions, by an explicit Grönwall bound in one direction and an explicit period-function obstruction in the other. We state four structural limits — Darboux triviality, the failure of the symplectic “category” to be a category, the Groenewold–van Hove obstruction, and symplectic rigidity — and give falsification conditions. We do not claim the compositional thesis is established, only that this reconstruction makes it precise enough to be attacked. A Haskell appendix renders the type skeleton; it is a typing, not a formal proof.

1 Introduction and epistemic conventions

1.1 The thesis under test

The PRINCIPIA PHYSICA pilot investigates the thesis that *physics is the study of empirically realized compositional mathematical structures*. This paper does not assume that thesis. It performs one bounded task on which the thesis can be tested: it reconstructs finite-dimensional Hamiltonian mechanics as a typed object, records exactly which of its features are mathematical and which are empirical, and reports where the compositional reading succeeds, where it is silent, and where it provably fails.

Three separations organise the paper. **(S1) Structure versus law:** the kinematic type (M, ω) and the dynamical law $\iota_{X_H}\omega = dH$ are logically independent data, and Section 3 shows what each contributes. **(S2) Mathematical versus physical equivalence:** an isomorphism of the underlying mathematical objects is neither necessary nor sufficient for empirical indistinguishability, and Section 5 proves both non-implications. **(S3) Composition versus interaction:** the monoidal product of Hamiltonian systems exists, and Section 4 proves that it cannot express interaction, so any compositional account of physics must supply an interconnection operation that is not $\otimes$.

1.2 Scope

Everything below is confined to smooth, finite-dimensional, real classical systems. Field theory, infinite-dimensional analysis, general relativity, quantum theory, and quantum field theory appear only as *boundary tests*: they are named to mark where the present type stops applying, and no result is claimed inside them. No new experimental claim is made anywhere in this paper.

1.3 Epistemic conventions

Every numbered statement carries exactly one of the following types, and the types share a single counter so that a statement’s number never encodes its epistemic status: *Definition*, *Axiom*, *Assumption*, *Conjecture*, *Lemma*, *Proposition*, *Theorem*, *Corollary*, *Interpretation*, *Empirical statement*, *Established result*, *New result claimed by this work*. We use them as follows. *Axiom* marks a stipulation constitutive of the object being defined; it is not asserted to be true of the world. *Assumption* marks a restriction adopted to make a proof go through; assumptions are never upgraded to theorems anywhere in this paper, and where one fails we give a non-example. *Established result* marks a statement proved elsewhere and cited, not reproved here. *New result claimed by this work* marks statements we prove here and put forward as the contribution; where a statement is elementary and probably folklore we say so in the same breath, and the claim is then to the *explicit statement* rather than to mathematical novelty. *Empirical statement* marks a claim about measurement outcomes, either cited to a measurement or flagged as an input. *Interpretation* marks a semantic

bridge between formal structure and physical vocabulary; interpretations are not derived. All references were checked for bibliographic identity before citation, and Appendix B records what was confirmed for each and what remains inferred.

2 The typed object

2.1 Types

Definition 2.1 (Type vocabulary). We use five sorts. A *kinematic type* is a pair (M, ω) with M a smooth manifold and ω a differential 2-form on M . An *observable type* over (M, ω) is the commutative $\mathbb{R}$ -algebra $\mathcal{A}(M) := C^\infty(M, \mathbb{R})$. A *generator* is an element $H \in \mathcal{A}(M)$. A *process type* over (M, ω) is a smooth local flow on M . An *interpretation* is the data specified in Definition 2.11.

Definition 2.2 (Hamiltonian theory object). A *Hamiltonian theory object* is a tuple

$$\mathbb{T} = (M, \omega, \mathcal{A}(M), H, \Phi, \mathcal{I})$$

satisfying Axioms 2.3–2.8 below. We write $\mathbb{T}^{\text{math}} = (M, \omega, \mathcal{A}(M), H, \Phi)$ for the *bare* object obtained by discarding the interpretation, and $U : \mathbb{T} \mapsto \mathbb{T}^{\text{math}}$ for that forgetting operation.

2.2 Axioms

Axiom 2.3 (Kinematic type). M is a smooth, second-countable, Hausdorff manifold of finite dimension $\dim M = 2n$ with $n \geq 0$. Every substantive statement below has $n \geq 1$; the case $n = 0$ is admitted solely so that the one-point object can serve as the monoidal unit in Corollary 4.9.

Axiom 2.4 (Symplectic structure). $\omega \in \Omega^2(M)$ satisfies $d\omega = 0$ and is nondegenerate: for each $x \in M$ the map $\omega_x^b : T_x M \rightarrow T_x^* M, v \mapsto \omega_x(v, \cdot)$, is a linear isomorphism.

Axiom 2.5 (Observable type). The observables are $\mathcal{A}(M) = C^\infty(M, \mathbb{R})$ with pointwise operations. Physical quantities are elements of $\mathcal{A}(M)$; the value of the quantity f in the state x is the real number $f(x)$.

Axiom 2.6 (Generator). A distinguished element $H \in \mathcal{A}(M)$ is given.

Axiom 2.7 (Dynamical law). The evolution Φ is the maximal local flow of the unique vector field X_H determined by $\iota_{X_H} \omega = dH$, and evolution is autonomous: $\Phi_{t+s} = \Phi_t \circ \Phi_s$ wherever both sides are defined.

Axiom 2.8 (Interpretation). An interpretation $\mathcal{I}$ in the sense of Definition 2.11 is given.

Axioms 2.3–2.8 are stipulations. Nothing in this paper asserts that any physical system satisfies them; Section 5 addresses when the assertion is warranted and Section 6 exhibits systems for which each of Axioms 2.3, 2.4 and 2.7 fails.

Assumption 2.9 (Completeness). Where a statement quantifies over all $t \in \mathbb{R}$ we assume X_H is complete, so that Φ is a global $\mathbb{R}$ -action. This assumption is used only where stated.

Assumption 2.9 is a genuine restriction, not a technicality.

Non-example 2.10 (Incompleteness). On $(\mathbb{R}^2, \omega_0 = dq \wedge dp)$ take $H(q, p) = \frac{1}{2}p^2 - q^4$. On the level set $H = E$ one has $\dot{q} = p = \pm\sqrt{2(E + q^4)}$ along the branch $p > 0$, so the time to reach $q = +\infty$ from q_0 with $E + q_0^4 > 0$ is

$$\int_{q_0}^{\infty} \frac{dq}{\sqrt{2(E + q^4)}} < \infty,$$

because the integrand is $O(q^{-2})$ as $q \rightarrow \infty$. Hence X_H is incomplete: the solution leaves every compact set in finite time. This object satisfies Axioms 2.3–2.7 and violates Assumption 2.9.

2.3 The interpretation

Definition 2.11 (Interpretation, empirical model). An *interpretation* of a bare object $\mathbb{T}^{\text{math}}$ is a tuple $\mathcal{I} = (P, \mu, \mathcal{U}, \varepsilon, T)$ where

- $P \subseteq M$ is the set of *preparable* states, the image of a laboratory preparation procedure;
- $\mu : M \rightarrow \mathbb{R}^k$ is a smooth *readout map*, the composite of the quantities actually recorded by the apparatus;
- $\mathcal{U}$ assigns a physical dimension and unit to each component of μ and to H ;
- $\varepsilon > 0$ is the readout resolution and $T > 0$ the accessible time window.

The *empirical model* of $\mathbb{T}$ is

$$\text{Emp}(\mathbb{T}) = \{ (z, t, \mu(\Phi_t z)) : z \in P, t \in [0, T] \} \subseteq P \times [0, T] \times \mathbb{R}^k,$$

recorded together with $\mathcal{U}$ and read to precision ε .

Interpretation 2.12 (Locus of empirical content). The empirical content of $\mathbb{T}$ resides in $\text{Emp}(\mathbb{T})$, not in $\mathbb{T}^{\text{math}}$. Two consequences are used below: (i) any feature of (M, ω, H) invisible to μ on P within ε over $[0, T]$ is, for that interpretation, empirically idle; (ii) $\mathcal{I}$ is not recoverable from $\mathbb{T}^{\text{math}}$, because U discards it, which is what Proposition 5.3 makes precise.

3 Reconstruction: what the axioms generate

3.1 The dynamical law is well posed

Lemma 3.1 (Existence and uniqueness of X_H). *Under Axioms 2.3–2.6 there is exactly one smooth vector field X_H on M with $\iota_{X_H}\omega = dH$.*

Proof. By Axiom 2.4 the bundle map $\omega^b : TM \rightarrow T^*M$ is a fibrewise linear isomorphism, and it is smooth because ω is. A smooth bundle isomorphism has a smooth inverse $(\omega^b)^{-1}$. Setting $X_H := (\omega^b)^{-1}(dH)$ gives a smooth vector field with $\iota_{X_H}\omega = dH$. If X and X' both satisfy the equation then $\iota_{X-X'}\omega = 0$, and nondegeneracy forces $X = X'$ pointwise. $\square$

Definition 3.2 (Poisson bracket). For $f, g \in \mathcal{A}(M)$ set $\{f, g\} := \omega(X_f, X_g)$.

Lemma 3.3 (Bracket identities). *For all $f, g, h \in \mathcal{A}(M)$:*

- (i) $\{f, g\} = X_g(f) = -\{g, f\}$;
- (ii) $\{fg, h\} = f\{g, h\} + g\{f, h\}$ (Leibniz rule);
- (iii) $\iota_{[X, Y]}\omega = d(\omega(Y, X))$ for all X, Y with $\mathcal{L}_X\omega = \mathcal{L}_Y\omega = 0$;
- (iv) $[X_f, X_g] = -X_{\{f, g\}}$;
- (v) $\{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\} = 0$.

Hence $(\mathcal{A}(M), \cdot, \{ \cdot, \cdot \})$ is a Poisson algebra.

Proof. (i) $\omega(X_f, X_g) = (\iota_{X_f}\omega)(X_g) = df(X_g) = X_g(f)$, and antisymmetry is that of ω . (ii) X_h is a derivation of $\mathcal{A}(M)$ and $\{f, h\} = X_h(f)$ by (i). (iii) Cartan's formula gives $\mathcal{L}_X = d\iota_X + \iota_X d$; with $d\omega = 0$ this yields $\mathcal{L}_X\omega = d\iota_X\omega$, so $\iota_X\omega$ and $\iota_Y\omega$ are closed. Using $\iota_{[X, Y]}\omega = \mathcal{L}_X\iota_Y\omega - \iota_Y(\mathcal{L}_X\omega) = (d\iota_X + \iota_X d)(\iota_Y\omega) - 0 = d(\iota_X\iota_Y\omega) = d(\omega(Y, X))$,

$$\iota_{[X, Y]}\omega = \mathcal{L}_X(\iota_Y\omega) - \iota_Y(\mathcal{L}_X\omega) = (d\iota_X + \iota_X d)(\iota_Y\omega) - 0 = d(\iota_X\iota_Y\omega) = d(\omega(Y, X)),$$

where $d(\iota_Y\omega) = 0$ was used. (iv) Every X_f satisfies $\mathcal{L}_{X_f}\omega = ddf = 0$, so (iii) applies and gives $\iota_{[X_f, X_g]}\omega = d(\omega(X_g, X_f)) = d\{g, f\} = -d\{f, g\} = -\iota_{X_{\{f, g\}}}\omega$; nondegeneracy gives the identity. (v) Using (i) and (iv),

$$\{f, \{g, h\}\} = X_{\{g, h\}}(f) = -[X_g, X_h](f) = -X_g(X_h(f)) + X_h(X_g(f)) = -\{\{f, h\}, g\} + \{g, \{f, h\}\}.$$

Antisymmetry, applied twice on the left factor and once on the right, gives $\{\{f, h\}, g\} = -\{g, \{f, h\}\} = \{g, \{h, f\}\}$ and $\{\{f, g\}, h\} = -\{h, \{f, g\}\}$. Substituting, $\{f, \{g, h\}\} = -\{g, \{h, f\}\} - \{h, \{f, g\}\}$, which is (v). $\square$

Remark 3.4 (Sign conventions are part of the type). Item (iv) carries a minus sign under the conventions $\iota_{X_H}\omega = dH$ and $\{f, g\} = \omega(X_f, X_g)$ fixed above, so $f \mapsto X_f$ is a Lie algebra *anti*-homomorphism and $f \mapsto -X_f$ is a homomorphism. A check on $(\mathbb{R}^2, dq \wedge dp)$ with $f = \frac{1}{2}q^2$, $g = \frac{1}{2}p^2$ gives $X_f = -q\partial_p$, $X_g = p\partial_q$, $[X_f, X_g] = -q\partial_q + p\partial_p$, and $\{f, g\} = qp$ with $X_{qp} = q\partial_q - p\partial_p$. A compositional reading that treats $f \mapsto X_f$ as a functor must carry this sign in the data.

Proposition 3.5 (Coordinate form; conservation; Liouville). *Let $(q^1, \dots, q^n, p_1, \dots, p_n)$ be local coordinates in which $\omega = \sum_i dq^i \wedge dp_i$. Then*

$$X_H = \sum_{i=1}^n \left(\frac{\partial H}{\partial p_i} \frac{\partial}{\partial q^i} - \frac{\partial H}{\partial q^i} \frac{\partial}{\partial p_i} \right), \quad \dot{q}^i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q^i},$$

$\{f, g\} = \sum_i (\partial_{q^i} f \partial_{p_i} g - \partial_{p_i} f \partial_{q^i} g)$, and moreover $X_H(H) = 0$ and $\mathcal{L}_{X_H}\omega = 0$, hence $\mathcal{L}_{X_H}\omega^{\wedge n} = 0$.

Proof. Write $X = \sum_i (A^i \partial_{q^i} + B_i \partial_{p_i})$. Then $\iota_X\omega = \sum_i (A^i dp_i - B_i dq^i)$. Equating with $dH = \sum_i (\partial_{q^i} H dq^i + \partial_{p_i} H dp_i)$ gives $A^i = \partial_{p_i} H$ and $B_i = -\partial_{q^i} H$, which is the displayed field and, applied to a curve, Hamilton's equations. The bracket formula is Lemma 3.3(i) in these coordinates. $X_H(H) = dH(X_H) = \omega(X_H, X_H) = 0$ by antisymmetry. $\mathcal{L}_{X_H}\omega = d\iota_{X_H}\omega + \iota_{X_H}d\omega = d dH + 0 = 0$, and the Lie derivative is a derivation of the wedge product, so the volume form $\omega^{\wedge n}$ is preserved. $\square$

Established result 3.6 (Darboux). Under Axioms 2.3–2.4, every point of M has a neighbourhood carrying coordinates in which $\omega = \sum_{i=1}^n dq^i \wedge dp_i$ [1, Ch. 3], [2, §43].

Theorem 3.7 (Local reconstruction). *Let $\mathbb{T}$ satisfy Axioms 2.3–2.7. Then every point of M has a neighbourhood V and coordinates on V in which the observable type is $C^\infty(V, \mathbb{R})$ with the canonical Poisson bracket, the dynamical law is Hamilton's equations for $H|_V$, and the flow preserves the canonical volume. Conversely, any system of first-order ODEs on an open $V \subseteq \mathbb{R}^{2n}$ that takes the form $\dot{q} = \partial_p H$, $\dot{p} = -\partial_q H$ for some $H \in C^\infty(V, \mathbb{R})$ arises from the object $(V, \sum_i dq^i \wedge dp_i, C^\infty(V, \mathbb{R}), H, \Phi)$ satisfying Axioms 2.3–2.7.*

Proof. The forward direction combines Established result 3.6 with Proposition 3.5. For the converse, $\omega_0 = \sum_i dq^i \wedge dp_i$ is closed (constant coefficients) and nondegenerate (its matrix in the frame ∂_q, ∂_p is the standard symplectic matrix, which is invertible), so Axioms 2.3–2.4 hold; Axiom 2.5 is a definition; H supplies Axiom 2.6; and the computation in Proposition 3.5, read backwards, identifies the given ODE with the flow equation of X_H , giving Axiom 2.7. $\square$

Remark 3.8 (What Theorem 3.7 does *not* say). It is a statement about local normal form. It does not assert that a physical system is described by such an object, and it supplies no criterion for deciding whether a given empirical dynamics admits a Hamiltonian presentation. That decision problem is the inverse problem of the calculus of variations, solved for two-dimensional second-order systems by Douglas [4] and open in general.

3.2 How much of H the dynamics determines

Theorem 3.9 (Exact sequence of the kinematic type). *Let $\mathfrak{X}_{\text{Symp}}(M) := \{X : \mathcal{L}_X \omega = 0\}$ and $\mathfrak{X}_{\text{Ham}}(M) := \{X_f : f \in \mathcal{A}(M)\}$. Then*

$$0 \longrightarrow H_{\text{dR}}^0(M) \longrightarrow \mathcal{A}(M) \xrightarrow{f \mapsto X_f} \mathfrak{X}_{\text{Symp}}(M) \xrightarrow{X \mapsto [\iota_X \omega]} H_{\text{dR}}^1(M) \longrightarrow 0$$

is an exact sequence of real vector spaces. Moreover $\mathfrak{X}_{\text{Ham}}(M)$ is a Lie ideal in $\mathfrak{X}_{\text{Symp}}(M)$ and $[\mathfrak{X}_{\text{Symp}}(M), \mathfrak{X}_{\text{Symp}}(M)] \subseteq \mathfrak{X}_{\text{Ham}}(M)$.

Proof. $H_{\text{dR}}^0(M)$ is the space of locally constant functions. If $X_f = 0$ then $df = \iota_{X_f} \omega = 0$, so f is locally constant; conversely a locally constant f has $df = 0$, hence $X_f = 0$ by Lemma 3.1. This gives exactness at $H_{\text{dR}}^0(M)$ and at $\mathcal{A}(M)$. Every X_f satisfies $\mathcal{L}_{X_f} \omega = d df = 0$, so the second map lands in $\mathfrak{X}_{\text{Symp}}(M)$. For $X \in \mathfrak{X}_{\text{Symp}}(M)$, Cartan's formula and $d\omega = 0$ give $d(\iota_X \omega) = \mathcal{L}_X \omega = 0$, so $[\iota_X \omega]$ is defined. The class vanishes exactly when $\iota_X \omega = df$ for some f , that is exactly when $X = X_f$; this is exactness at $\mathfrak{X}_{\text{Symp}}(M)$. For surjectivity onto $H_{\text{dR}}^1(M)$: given a closed 1-form α , put $X := (\omega^\flat)^{-1} \alpha$; then $\iota_X \omega = \alpha$ is closed, so $\mathcal{L}_X \omega = d\alpha = 0$ and $X \in \mathfrak{X}_{\text{Symp}}(M)$ with $[\iota_X \omega] = [\alpha]$. The last claim is Lemma 3.3(iii): for $X, Y \in \mathfrak{X}_{\text{Symp}}(M)$ the form $\iota_{[X, Y]} \omega = d(\omega(Y, X))$ is exact, so $[X, Y] = X_g$ with $g := \omega(Y, X)$ by Lemma 3.1; in particular $[X, Y] \in \mathfrak{X}_{\text{Ham}}(M)$, and taking $Y \in \mathfrak{X}_{\text{Ham}}(M)$ shows $\mathfrak{X}_{\text{Ham}}(M)$ is an ideal. $\square$

Corollary 3.10 (Underdetermination of the generator). *If M is connected, the flow Φ determines H up to one additive real constant. If $H_{\text{dR}}^1(M) \neq 0$, there exist ω -preserving flows that are generated by no global H whatsoever.*

Proof. The first claim is exactness at $\mathcal{A}(M)$ together with $H_{\text{dR}}^0(M) \cong \mathbb{R}$ for connected M : two generators of the same field differ by a locally constant, hence constant, function. The second is exactness at $\mathfrak{X}_{\text{Symp}}(M)$: any X with $[\iota_X \omega] \neq 0$ preserves ω and is not of the form X_f . $\square$

Non-example 3.11 (Symplectic but not Hamiltonian). Let $M = \mathbb{T}^2 = \mathbb{R}^2 / (2\pi\mathbb{Z})^2$ with angle coordinates (θ^1, θ^2) and $\omega = d\theta^1 \wedge d\theta^2$, which is closed and nondegenerate. Let $X = \partial_{\theta^1}$. Then $\iota_X \omega = d\theta^2$, which is closed but not exact, since $\int_\gamma d\theta^2 = 2\pi \neq 0$ for γ the second generating circle whereas the integral of an exact form over a loop vanishes. Hence X preserves ω and the Liouville area, its flow is a one-parameter group of symplectomorphisms, and by Corollary 3.10 no $H \in C^\infty(\mathbb{T}^2)$ has $X = X_H$. Axiom 2.7 therefore excludes a dynamics that satisfies every conservation property usually cited as the physical content of Hamiltonian mechanics.

Interpretation 3.12 (Reading of Corollary 3.10). The additive constant in H is invisible to $\text{Emp}(\mathbb{T})$ whenever the readout μ is built from the flow, because H and $H + c$ generate the same X_H by Lemma 3.1. Within the scope of this paper, an absolute zero of energy is therefore not an empirically realized element of the theory object. That classical gravitation and general relativity change this situation is noted as a boundary fact and is out of scope.

3.3 Symmetry

Proposition 3.13 (Hamiltonian reciprocity, and conservation). *For $F, H \in \mathcal{A}(M)$ the following are equivalent: (a) $\{F, H\} = 0$; (b) F is constant along the flow of X_H ; (c) H is constant along the flow of X_F .*

Proof. By Lemma 3.3(i), $\frac{d}{dt}(F \circ \Phi_t^H) = X_H(F) \circ \Phi_t^H = \{F, H\} \circ \Phi_t^H$, giving (a) $\Leftrightarrow$ (b) because a smooth function on a connected time interval is constant exactly when its derivative vanishes identically. Likewise $\frac{d}{dt}(H \circ \Phi_t^F) = \{H, F\} \circ \Phi_t^F$, and $\{H, F\} = -\{F, H\}$, giving (a) $\Leftrightarrow$ (c). $\square$

The group-theoretic strengthening — for a Lie group G acting by symplectomorphisms with an Ad^* -equivariant momentum map $J : M \rightarrow \mathfrak{g}^*$ and G -invariant H , each component of J is conserved — is an established result we use only by citation [9, Chs. 11–13], [1, Ch. 4].

4 Composition, and what composition cannot do

4.1 Morphisms

Definition 4.1 (Morphism, isomorphism). A *morphism* $\varphi : (M, \omega, H) \rightarrow (M', \omega', H')$ of bare objects is a smooth map $\varphi : M \rightarrow M'$ with $\varphi^* \omega' = \omega$ and $H = H' \circ \varphi + c$ for some constant $c \in \mathbb{R}$. It is an *isomorphism* when φ is in addition a diffeomorphism. An *isomorphism of theory objects* is an isomorphism of bare objects together with a bijection of interpretations $\mathcal{I} \rightarrow \mathcal{I}'$ intertwining $P, \mu, \mathcal{U}$.

Lemma 4.2 (Symplectic maps are immersions). *If $\varphi^* \omega' = \omega$ with ω nondegenerate, then $d\varphi_x$ is injective for every $x \in M$.*

Proof. Suppose $d\varphi_x(v) = 0$. For every $w \in T_x M$, $\omega_x(v, w) = (\varphi^* \omega')_x(v, w) = \omega'_{\varphi(x)}(d\varphi_x v, d\varphi_x w) = 0$. Nondegeneracy of ω_x gives $v = 0$. $\square$

New result claimed by this work 4.3 (No symplectic coarse-graining). There is no morphism $\varphi : (M, \omega, H) \rightarrow (M', \omega', H')$ in the sense of Definition 4.1 with $\dim M > \dim M'$. In particular, projecting a many-degree-of-freedom object onto a few-degree-of-freedom object is never a morphism of Hamiltonian theory objects.

Proof. By Lemma 4.2, $d\varphi_x$ is injective, so $\dim M = \dim T_x M \leq \dim T_{\varphi(x)} M' = \dim M'$. $\square$

Remark 4.4. Lemma 4.2 is standard and New result 4.3 is elementary, so we expect the latter is folklore; the claim here is to its *explicit statement in the typed setting*, because it settles a question a compositional account must answer. Coarse-graining, thermodynamic reduction and effective-model extraction are *not* morphisms in this category. They are quotient constructions — symplectic reduction [9] — so a compositional framework that wants them must carry a second class of arrows, or work in a double category or a category of spans.

Lemma 4.5 (Isomorphisms intertwine flows). *If φ is an isomorphism then $d\varphi \circ X_H = X_{H'} \circ \varphi$ and $\varphi \circ \Phi_t = \Phi'_t \circ \varphi$ on the common domain of definition.*

Proof. For $w \in T_x M$, $\omega'_{\varphi(x)}(d\varphi X_H, d\varphi w) = (\varphi^* \omega')_x(X_H, w) = \omega_x(X_H, w) = dH_x(w) = d(H' \circ \varphi)_x(w) = dH'_{\varphi(x)}(d\varphi w)$. As φ is a diffeomorphism, $d\varphi_x$ is onto $T_{\varphi(x)} M'$, so $\iota_{d\varphi X_H} \omega' = dH'$ at $\varphi(x)$, and Lemma 3.1 gives $d\varphi X_H = X_{H'} \circ \varphi$. Two vector fields that are φ -related have φ -related flows. $\square$

4.2 The monoidal product and its exact range

Definition 4.6 (Cartesian product of objects). For bare objects $\mathbb{T}_i^{\text{math}} = (M_i, \omega_i, \mathcal{A}(M_i), H_i, \Phi^i)$, $i = 1, 2$, set

$$\mathbb{T}_1^{\text{math}} \otimes \mathbb{T}_2^{\text{math}} := (M_1 \times M_2, \pi_1^* \omega_1 + \pi_2^* \omega_2, \mathcal{A}(M_1 \times M_2), H_1 \circ \pi_1 + H_2 \circ \pi_2, \Phi).$$

Proposition 4.7 ($\otimes$ is well typed and non-interacting). $\omega := \pi_1^* \omega_1 + \pi_2^* \omega_2$ satisfies Axiom 2.4 on $M_1 \times M_2$, and with $H_\otimes := H_1 \circ \pi_1 + H_2 \circ \pi_2$ one has $X_{H_\otimes} = X_{H_1} \oplus X_{H_2}$ and $\Phi_t = \Phi_t^1 \times \Phi_t^2$.

Proof. $d\omega = \pi_1^* d\omega_1 + \pi_2^* d\omega_2 = 0$. Under the canonical splitting $T_{(x_1, x_2)}(M_1 \times M_2) \cong T_{x_1} M_1 \oplus T_{x_2} M_2$ the form ω is the block-diagonal pairing $\omega_1 \oplus \omega_2$, and a block-diagonal bilinear form with nondegenerate blocks is nondegenerate: if $\omega((v_1, v_2), (w_1, w_2)) = \omega_1(v_1, w_1) + \omega_2(v_2, w_2)$ vanishes for all (w_1, w_2) , then taking $w_2 = 0$ gives $v_1 = 0$ and taking $w_1 = 0$ gives $v_2 = 0$. For the vector field, with $X := X_{H_1} \oplus X_{H_2}$, $\iota_X \omega = \pi_1^*(\iota_{X_{H_1}} \omega_1) + \pi_2^*(\iota_{X_{H_2}} \omega_2) = \pi_1^* dH_1 + \pi_2^* dH_2 = dH_\otimes$, and Lemma 3.1 identifies $X = X_{H_\otimes}$. The flow of a direct-sum field on a product is the product of the flows. $\square$

New result claimed by this work 4.8 (Interaction is outside the range of $\otimes$). Let $M = M_1 \times M_2$ carry ω and $H_\otimes$ as above and let $H = H_\otimes + V$ with $V \in \mathcal{A}(M)$. Then the following are equivalent:

- (a) $\Phi_t^H = \Phi_t^1 \times \Phi_t^2$ for all t in a neighbourhood of 0 and all points of M ;
- (b) $X_V = 0$;
- (c) V is locally constant.

Consequently the image of $\otimes$ inside the class of objects on $M_1 \times M_2$ with the split symplectic form consists exactly of the composites whose factor dynamics are independent; no choice of factors makes $\otimes$ produce an interacting composite.

Proof. (c) $\Rightarrow$ (b): $dV = 0$ gives $X_V = (\omega^\flat)^{-1}(0) = 0$. (b) $\Rightarrow$ (a): $X_H = X_{H_\otimes} + X_V$ by linearity of $f \mapsto X_f$ (Lemma 3.1 and linearity of d and of $(\omega^\flat)^{-1}$); with $X_V = 0$, Proposition 4.7 gives the product flow. (a) $\Rightarrow$ (b): differentiating $\Phi_t^H(x)$ and $(\Phi_t^1 \times \Phi_t^2)(x)$ at $t = 0$ gives $X_H(x) = (X_{H_1} \oplus X_{H_2})(x) = X_{H_\otimes}(x)$ for every x , hence $X_V = X_H - X_{H_\otimes} = 0$. (b) $\Rightarrow$ (c): $X_V = 0$ gives $\iota_{X_V} \omega = dV = 0$ by Lemma 3.1, so V is locally constant. For the final sentence: an object on $(M_1 \times M_2, \omega)$ with generator H lies in the image of $\otimes$ exactly when $H - H_\otimes$ is locally constant for some H_1, H_2 , which by the equivalence is exactly when its flow splits. $\square$

Corollary 4.9 (A compositional account needs a second operation). *Under Definition 4.6, the assignment $(\mathbb{T}_1, \mathbb{T}_2) \mapsto \mathbb{T}_1 \otimes \mathbb{T}_2$ is a symmetric monoidal product on bare objects with unit the one-point object $(\{*\}, 0, \mathbb{R}, 0)$, and by New result 4.8 every interacting system on $M_1 \times M_2$ is outside its range. Hence no account of physical composition that uses only $\otimes$ can express interaction.*

Proof. Symmetry and associativity up to the canonical diffeomorphisms of products are inherited from the Cartesian product of manifolds; each canonical diffeomorphism pulls back the split form to the split form and matches the generators, so it is an isomorphism in the sense of Definition 4.1. Degeneracy of ω is not an issue for the unit because $\dim\{*\} = 0$ and the empty pairing is vacuously nondegenerate. The second sentence is New result 4.8. Combining gives the conclusion. $\square$

Remark 4.10 (Relation to the applied-category-theory literature). Corollary 4.9 is the precise form of a widely acknowledged point: interconnection is modelled by cospans, decorated cospans, or relations, not by monoidal product alone [5, 3, 8]. Our contribution here is only the sharp “if and only if” of New result 4.8 inside the typed Hamiltonian setting, which converts a methodological preference into a theorem about what $\otimes$ can represent.

4.3 The natural repair fails to be a category

Established result 4.11 (The symplectic “category” is not a category). Take symplectic manifolds as objects and Lagrangian submanifolds of $M_1^- \times M_2$ (canonical relations) as arrows. The set-theoretic composite of two canonical relations need not be a submanifold, and when it is, need not be Lagrangian; composability requires transversality and clean-intersection hypotheses. Consequently this data does not form a category without further work; see Weinstein [10] for the statement of the difficulty and for the standard repairs.

Interpretation 4.12 (Cost of Established result 4.11). The natural candidate for a composition rich enough to carry interaction — composition of canonical relations — is only *partially* defined. A compositional foundation for classical mechanics must therefore accept a partial composition, pass to a completion, or restrict to a subclass on which transversality is guaranteed. This is a structural limit on the thesis under test, not a gap in exposition.

5 Empirical realization, equivalence, and limits

5.1 Two directions of a single separation

Definition 5.1 (ε -empirical equivalence). Let $\mathbb{T}, \mathbb{T}'$ be theory objects with the same M , the same preparable set P , the same readout μ and units, resolution ε , and window $[0, T]$. They are ε -empirically equivalent on (P, T) when

$$\sup_{z \in P} \sup_{t \in [0, T]} \|\mu(\Phi_t z) - \mu(\Phi'_t z)\| \leq \varepsilon.$$

For objects on different manifolds the comparison requires an explicit translation of preparations and readouts; we do not use that case.

Proposition 5.2 (Empirically equivalent, mathematically non-isomorphic). *Fix $m, k > 0$, $R > 0$, $T > 0$, and on $(\mathbb{R}^2, \omega_0 = dq \wedge dp)$ set*

$$H_0 = \frac{p^2}{2m} + \frac{k}{2}q^2, \quad H_\lambda = H_0 + \lambda q^4 \quad (0 < \lambda \leq 1), \quad P = \{z : \|z\| \leq R\}.$$

Then:

- (i) *there is $C = C(m, k, R, T) < \infty$, independent of λ , with $\sup_{z \in P} \sup_{t \in [0, T]} \|\Phi_t^\lambda z - \Phi_t^0 z\| \leq C\lambda$; hence for every $\varepsilon > 0$ and every 1-Lipschitz readout μ there is $\lambda > 0$ making $\mathbb{T}_0$ and $\mathbb{T}_\lambda$ ε -empirically equivalent on (P, T) ;*
- (ii) *for every $\lambda > 0$ the bare objects $(\mathbb{R}^2, \omega_0, H_0)$ and $(\mathbb{R}^2, \omega_0, H_\lambda)$ are not isomorphic in the sense of Definition 4.1.*

Proof. (i) Put $E_R := \sup_{\|z\| \leq R} H_1(z) < \infty$. For $\lambda \in (0, 1]$ and $z \in P$ one has $H_\lambda(z) \leq H_1(z) \leq E_R$, and H_λ is conserved along Φ^λ by Proposition 3.5. Since $H_0 \leq H_\lambda$ pointwise, the trajectory satisfies $H_0(\Phi_t^\lambda z) \leq E_R$, so it remains in the compact set $K := \{H_0 \leq E_R\}$; the same bound holds for $\lambda = 0$. Set $q_{\max} := \sqrt{2E_R/k}$ and $A_R := 4q_{\max}^3$, an upper bound on $\|X_{q^4}\|$ on K (indeed $X_{q^4} = -4q^3 \partial_p$). The field $X_{H_0}(q, p) = (p/m, -kq)$ is linear, hence globally Lipschitz with some constant $L > 0$. Writing $u(t) := \Phi_t^\lambda z - \Phi_t^0 z$,

$$\dot{u} = [X_{H_0}(\Phi_t^\lambda z) - X_{H_0}(\Phi_t^0 z)] + \lambda X_{q^4}(\Phi_t^\lambda z), \quad \| \dot{u} \| \leq L \| u \| + \lambda A_R, \quad u(0) = 0.$$

Grönwall’s inequality gives $\|u(t)\| \leq (\lambda A_R/L)(e^{Lt} - 1)$, so $C := (A_R/L)(e^{LT} - 1)$ works. A 1-Lipschitz μ transports the bound, and $\lambda < \varepsilon/C$ gives the equivalence.

(ii) For $\lambda \geq 0$ the potential $V_\lambda(q) = \frac{k}{2}q^2 + \lambda q^4$ is smooth, even, strictly increasing on $[0, \infty)$ with $V_\lambda(q) \rightarrow \infty$, and $V'_\lambda(q) = kq + 4\lambda q^3$ vanishes only at $q = 0$. Hence for every $E > 0$ the level set $\{H_\lambda = E\}$ is a regular, compact, connected curve, and the orbit on it is periodic with period $T_\lambda(E)$. Let $A_\lambda(E)$ be the area enclosed. The classical relation $T_\lambda(E) = A'_\lambda(E)$ for one-degree-of-freedom systems with regular compact level sets [2, Ch. 10] is used.

For $\lambda = 0$ the level set is the ellipse with semi-axes $\sqrt{2E/k}$ and $\sqrt{2mE}$, so $A_0(E) = 2\pi E\sqrt{m/k}$ and $T_0(E) = 2\pi\sqrt{m/k}$ for every $E > 0$: the period function is constant.

For $\lambda > 0$ suppose, for contradiction, that $T_\lambda \equiv c$ on $(0, \infty)$ for some $c > 0$. Since A_λ is continuous with $A_\lambda(0) = 0$ and $A'_\lambda = T_\lambda$, integration gives $A_\lambda(E) = cE$ for all $E \geq 0$. On the other hand $H_\lambda \geq \frac{p^2}{2m} + \lambda q^4$ pointwise, so $\{H_\lambda \leq E\} \subseteq \{\frac{p^2}{2m} + \lambda q^4 \leq E\}$ and

$$A_\lambda(E) \leq \int_{-\sqrt{2mE}}^{\sqrt{2mE}} 2\left(\frac{E - \frac{p^2}{2m}}{\lambda}\right)^{1/4} dp = 2\lambda^{-1/4}\sqrt{2mE} E^{1/4} \int_{-1}^1 (1 - u^2)^{1/4} du = \kappa \sqrt{m} \lambda^{-1/4} E^{3/4},$$

with $\kappa := 2\sqrt{2} \int_{-1}^1 (1 - u^2)^{1/4} du \in (0, \infty)$, using the substitution $p = \sqrt{2mE} u$. Combining, $cE \leq \kappa \sqrt{m} \lambda^{-1/4} E^{3/4}$, i.e. $E^{1/4} \leq \kappa \sqrt{m}/(c\lambda^{1/4})$ for all $E > 0$, which fails for E large. Hence T_λ is not constant.

Now suppose φ were an isomorphism $(\mathbb{R}^2, \omega_0, H_0) \rightarrow (\mathbb{R}^2, \omega_0, H_\lambda)$. By Lemma 4.5 it carries orbits of Φ^0 to orbits of Φ^λ preserving the time parameter, so it carries periodic orbits to periodic orbits of the same period, and it is onto. Every periodic orbit of Φ^λ with energy $E > 0$ is therefore the image of a periodic orbit of Φ^0 , whose period is $2\pi\sqrt{m/k}$. That makes T_λ constant on $(0, \infty)$, contrary to the previous paragraph. $\square$

Proposition 5.3 (Mathematically isomorphic, physically distinct). *There exist theory objects $\mathbb{T}, \mathbb{T}'$ with $U\mathbb{T} \cong U\mathbb{T}'$ as bare objects whose empirical models differ, and no isomorphism of bare objects induces a map $\text{Emp}(\mathbb{T}) \rightarrow \text{Emp}(\mathbb{T}')$.*

Proof. Take $\mathbb{T}$ the mass–spring object on $(\mathbb{R}^2, dq \wedge dp)$ with $H = p^2/2m + kq^2/2$, readout $\mu(q, p) = q$ with $\mathcal{U}(\mu)$ the metre and $\mathcal{U}(H)$ the joule, and P the states reachable by displacing the mass. Take $\mathbb{T}'$ the ideal LC circuit on $(\mathbb{R}^2, dQ \wedge d\Phi_f)$ with $H' = \Phi_f^2/2L + Q^2/2C$, readout $\mu'(Q, \Phi_f) = Q$ with $\mathcal{U}(\mu')$ the coulomb, and P' the states reachable by charging the capacitor. Choosing $m = L$ and $k = 1/C$, the identity map of $\mathbb{R}^2$ is an isomorphism of bare objects: it pulls back the form to the form and matches the generators exactly.

The empirical models differ as sets-with-units: $\text{Emp}(\mathbb{T})$ is a set of position records in metres and $\text{Emp}(\mathbb{T}')$ a set of charge records in coulombs, and $\mathcal{U}$ is part of the data of an interpretation by Definition 2.11. A bare isomorphism is a map of manifolds and carries no component acting on $\mathcal{U}$, because U deletes $\mathcal{U}$; hence it induces no map of empirical models. Two further differences are recorded as empirical inputs rather than derived here: the admissible interventions differ (one may clamp the spring; one may open the circuit), and the failure modes differ (plastic deformation; dielectric breakdown). $\square$

Interpretation 5.4 (The separation, stated). Propositions 5.2 and 5.3 together show that mathematical isomorphism of bare Hamiltonian objects is neither sufficient (5.3) nor necessary (5.2) for empirical indistinguishability. Any claim of the form “these two theories are the same physics because the mathematics is the same” therefore requires a separate empirical argument about $P, \mu, \mathcal{U}, \varepsilon$ and T . We adopt this as a working discipline rather than as a discovery: the pair of counterexamples is what makes the discipline enforceable.

5.2 Limiting relations

Example 5.5 (Small-oscillation limit is not an isomorphism). Let $\mathbb{T}^{\text{pend}}$ be the planar pendulum on T^*S^1 with $H^{\text{pend}}(\theta, p_\theta) = \frac{p_\theta^2}{2m\ell^2} + mg\ell(1 - \cos \theta)$, and let $\mathbb{T}^{\text{ho}}$ be the harmonic object on $T^*\mathbb{R}$ with $H^{\text{ho}} = \frac{p^2}{2m\ell^2} + \frac{1}{2}mg\ell\theta^2$. Since $1 - \cos \theta = \frac{1}{2}\theta^2 + O(\theta^4)$, on $\{|\theta| \leq \delta, |p_\theta| \leq \delta\}$ the two generators differ by $O(\delta^4)$ and a Grönwall estimate of the form used in Proposition 5.2(i) bounds the trajectory difference on $[0, T]$ by $C(T)\delta^4$. The two objects are not isomorphic: the pendulum has a non-constant period function and, unlike the harmonic object, has unbounded (θ -circulating) orbits and a separatrix. The relation between them is an *asymptotic* one, indexed by δ , not an isomorphism of theory objects.

Interpretation 5.6 (Reading of Example 5.5). Limiting relations between theory objects are relations between *interpreted* objects at stated resolution and time window; they are not recovered by any operation on bare objects alone, because the statement “ $C(T)\delta^4 \leq \varepsilon$ ” quantifies over ε and T , which live in $\mathcal{I}$.

5.3 An empirical anchor and an empirical refutation

Empirical statement 5.7 (Anchor). For laboratory mass–spring and LC systems in regimes where damping, non-linearity and drift are below the readout resolution, the object of Proposition 5.3 reproduces the recorded time series within ε over $[0, T]$. This is taken as an empirical input to the pilot; no new measurement is reported in this paper.

Established result 5.8 (Refutation of a specific Hamiltonian object). The two-body Newtonian gravitational object on $T^*(\mathbb{R}^3 \setminus \{0\})$ with $H = \frac{\|\mathbf{p}\|^2}{2\mu} - \frac{GM\mu}{\|\mathbf{q}\|}$ predicts a closed, non-precessing ellipse. The observed perihelion advance of Mercury exceeds the Newtonian N -body accounting by an amount matched by the general-relativistic prediction of 42.98'' per century [11]. The Hamiltonian object above is therefore empirically false at that precision.

Interpretation 5.9 (Reading of Established result 5.8). Established result 5.8 refutes one *object*, not the *type*: the first post-Newtonian correction is itself standardly presented by a Hamiltonian on a finite-dimensional phase space, so the

failure was of the generator H and not of Axioms 2.3–2.7. This distinction — refuting a member versus refuting a type — is what the typed presentation is for; Section 7 states what would refute the type.

6 Non-examples: where each axiom bites

Non-example 6.1 (Odd dimension: Axiom 2.3 bites). No manifold of odd dimension admits a nondegenerate 2-form. If $\dim V = d$ is odd and β is an alternating bilinear form on V , its matrix B in any basis satisfies $B^T = -B$, so $\det B = \det(-B^T) = (-1)^d \det B = -\det B$, giving $\det B = 0$ and hence a nonzero v with $\beta(v, \cdot) = 0$. Systems whose natural state space is odd-dimensional — for instance a state space carrying a contact structure, or a thermodynamic state space with an extra scalar — fall outside Axiom 2.3 and require a different kinematic type.

Non-example 6.2 (Degenerate bracket: Axiom 2.4 bites). On $\mathfrak{so}(3)^* \cong \mathbb{R}^3$ the Lie–Poisson bracket $\{f, g\}(\mathbf{x}) = -\mathbf{x} \cdot (\nabla f \times \nabla g)$ is a Poisson bracket for which the free rigid body, $H(\mathbf{x}) = \frac{1}{2} \sum_i x_i^2 / I_i$, generates Euler’s equations [9]. The associated bivector has rank 2 at $\mathbf{x} \neq 0$ and rank 0 at the origin, so it is not induced by any nondegenerate 2-form; the function $\|\mathbf{x}\|^2$ is a Casimir, i.e. $\{\|\mathbf{x}\|^2, \cdot\} = 0$, which is impossible for a nonconstant function under Axiom 2.4 because $\{f, \cdot\} = 0$ with $\{f, g\} = X_g(f) = -X_f(g)$ for all g forces $X_f = 0$, hence $df = 0$. The object is Poisson but not symplectic; the theory type must be widened, and widening it changes which statements of Section 3 survive.

New result claimed by this work 6.3 (Dissipation: Axiom 2.7 bites, for every ω). Let $\gamma > 0$, $\omega_0^2 > 0$ and let X be the damped-oscillator field on $\mathbb{R}^2$,

$$X(x, v) = (v, -\omega_0^2 x - \gamma v).$$

Then there is *no* symplectic form σ on $\mathbb{R}^2$ and no $H \in C^\infty(\mathbb{R}^2)$ with $\iota_X \sigma = dH$. In particular the obstruction is not an artefact of the standard form $dx \wedge dv$.

Proof. The linear part of X has matrix $\begin{pmatrix} 0 & 1 \\ -\omega_0^2 & -\gamma \end{pmatrix}$ with characteristic polynomial $s^2 + \gamma s + \omega_0^2$, whose roots have real part $-\gamma/2 < 0$ (for real roots, both are negative because their sum is $-\gamma < 0$ and product $\omega_0^2 > 0$). As X is linear, every solution is a combination of $e^{s \pm i t}$ modes, so $\Phi_t(z) \rightarrow 0$ as $t \rightarrow +\infty$ for every $z \in \mathbb{R}^2$.

Suppose $\iota_X \sigma = dH$ for some symplectic σ and some smooth H . Then $X(H) = dH(X) = \sigma(X, X) = 0$ by antisymmetry, so H is constant along every trajectory: $H(\Phi_t z) = H(z)$ for all $t \geq 0$. Letting $t \rightarrow +\infty$ and using continuity of H gives $H(z) = H(0)$ for every $z \in \mathbb{R}^2$. Hence $dH \equiv 0$, so $\iota_X \sigma \equiv 0$, and nondegeneracy of σ gives $X \equiv 0$. This contradicts $X(1, 0) = (0, -\omega_0^2) \neq 0$. $\square$

Remark 6.4 (Precision of New result 6.3). Three qualifications keep the statement honest. (a) The proof uses global asymptotic stability, hence $\gamma > 0$, $\omega_0^2 > 0$ and the global domain $\mathbb{R}^2$; on a proper subdomain avoiding the attractor, or with a *time-dependent* structure, Hamiltonian presentations of damped systems exist and are standard. (b) Being Hamiltonian is thus a property of the pair (X, σ) together with the domain, not of X alone; which second-order systems admit a variational or Hamiltonian presentation is the inverse problem of the calculus of variations [4]. (c) The result is elementary and the argument is likely known; the claim is to the explicit form-independent statement, which the typed setting requires and which is stronger than the usual divergence argument — the latter excludes only the standard ω_0 , since $\operatorname{div} X = -\gamma \neq 0$.

Established result 6.5 (No functorial quantization: the type does not extend). There is no map from the Poisson algebra of polynomials on $\mathbb{R}^{2n}$ to self-adjoint operators on a Hilbert space that is linear, sends 1 to the identity, sends $\{f, g\}$ to $\frac{1}{i\hbar} [\hat{f}, \hat{g}]$, and acts irreducibly on the linear polynomials. This is the Groenewold–van Hove obstruction; statements, proofs, and the analogous results for S^2 and T^*S^1 are collected by Gotay [6].

Interpretation 6.6 (Reading of Established result 6.5). A compositional programme that seeks a functor from classical theory objects to quantum theory objects preserving the full observable algebra structure cannot succeed on $\mathbb{R}^{2n}$: the obstruction is a theorem, not a missing construction. Quantum mechanics is a boundary test for this paper, and we claim nothing inside it; we record the obstruction only because it bounds what the reconstruction can be extended to.

Established result 6.7 (Symplectic rigidity). If $r > R$ there is no symplectic embedding of the ball $B^{2n}(r)$ into the cylinder $B^2(R) \times \mathbb{R}^{2n-2}$, although a volume-preserving embedding exists for all $r, R > 0$ when $n \geq 2$ [7].

Interpretation 6.8 (Reading of Established result 6.7). Axiom 2.4 is strictly stronger than “the flow preserves phase volume”. The kinematic type therefore carries information that the measure type does not, and the choice between them is in principle a substantive choice rather than a notational one.

Conjecture 6.9 (Empirical bite of rigidity). There exist a finite-dimensional classical laboratory system, an interpretation $\mathcal{I} = (P, \mu, \mathcal{U}, \varepsilon, T)$, and a pair of candidate objects agreeing on all volume-theoretic predictions within ε on (P, T) whose predictions for μ differ by more than ε solely as a consequence of Established result 6.7. We know of no such realization and do not assert that one exists.

Interpretation 6.10 (Reading of Established result 3.6). Established result 3.6 entails that symplectic manifolds of equal dimension have no local invariants: any two are locally isomorphic. Hence all local empirical content of

a Hamiltonian theory object resides in H and in $\mathcal{I}$, never in (M, ω) ; the kinematic type contributes only global information (topology of M , cohomology classes as in Theorem 3.9 and Non-example 3.11) and the dimension count. A compositional account that locates physical content in the kinematic type must therefore locate it globally, or the content is empty.

7 Falsification conditions and status of the thesis

Definition 7.1 (What would refute the type). We say the Hamiltonian type is *refuted for a domain D* of physical systems when there is a system in D , an interpretation $\mathcal{I}$, and recorded data $\mathcal{D}$ such that no object satisfying Axioms 2.3–2.7 on any finite-dimensional M has $\mathcal{D} \subseteq_{\varepsilon} \text{Emp}(\mathbb{T})$.

Three concrete refuting observations, each of which is currently unavailable and each of which is stated as a target rather than a claim:

- (F1) **Bracket failure.** Measured response functions for a closed finite-dimensional system whose induced bracket on the measured observables violates the Jacobi identity beyond ε . By Lemma 3.3(v) this refutes $d\omega = 0$, hence Axiom 2.4, for that system.
- (F2) **Global generator failure.** A closed finite-dimensional system whose measured evolution preserves ω but for which the class $[\iota_X \omega] \in H_{\text{dR}}^1(M)$ is measurably nonzero. By Corollary 3.10 and Non-example 3.11 this refutes Axiom 2.7 while leaving Axioms 2.3–2.5 intact — a discriminating rather than a global refutation.
- (F3) **Irreducible dissipation.** A system that is closed under the operative notion of isolation, is finite-dimensional, and exhibits global asymptotic stability of an equilibrium. By New result 6.3 this refutes Axiom 2.7 for that system under *every* symplectic form on its state space.

Interpretation 7.2 (Status of the pilot thesis after this paper). The reconstruction supports the following bounded report, and nothing stronger. *Precision:* the thesis can be stated precisely for this domain, by Definition 2.2, Axioms 2.3–2.8 and Definition 2.11. *Internal coherence:* the object is consistent and generates the standard formalism (Theorems 3.7 and 3.9). *Compatibility:* it reproduces the established structural results cited here without modification. *Usefulness:* the typing did work — it produced New results 4.3, 4.8 and 6.3, and forced the two-directional separation of Interpretation 5.4. *Costs:* the compositional reading is incomplete in a specific, provable way. $\otimes$ cannot express interaction (New result 4.8); the natural richer composition is only partial (Established result 4.11); the category has no coarse-graining arrows (New result 4.3); and the type does not extend functorially to quantum theory (Established result 6.5). We therefore do not report the thesis as established. We report that, for finite-dimensional Hamiltonian mechanics, it is precise, non-trivial, and partially falsifiable, with four named structural deficits.

Conjecture 7.3 (Interconnection completeness). There is a symmetric monoidal double category whose loose arrows are partially-defined canonical relations and whose tight arrows are symplectomorphisms, in which every object $(M_1 \times M_2, \omega_1 \oplus \omega_2, H_1 \oplus H_2 + V)$ arises as a composite of $\mathbb{T}_1, \mathbb{T}_2$ and one arrow determined by V . Decorated cospans [5] and the monoidal apparatus of [8, 3] are the natural tools; the obstruction to a naive construction is Established result 4.11.

A Type skeleton in Haskell

The following is a rendering of Definitions 2.1, 2.2, 2.11 and 4.1 as types. It is a *typing*, not a formal proof: it makes the arities and dependencies of the theory object machine-checkable, and it makes New result 4.8 expressible as a property, but nothing here verifies any theorem of this paper. No claim of formal verification is made.

```
module Principia.Hamiltonian where
type Tangent = [Double]; type Bilinear = (Tangent, Tangent) -> Double
type Observable m = m -> Double -- observable type
type Flow m = Double -> m -> Maybe m -- Maybe encodes the partiality

-- Kinematic type (kinematic + symplectic axioms); 'm' = points of M.
data Kinematic m = Kinematic { dimHalf :: Int, symplForm :: m -> Bilinear }
newtype Generator m = Generator (Observable m)

-- Interpretation. Units are DATA, not derivable from the bare object;
-- that is exactly what the "isomorphic but distinct" result turns on.
data Unit = Metre | Coulomb | Joule | Second | Other String deriving (Eq, Show)
data Interp m = Interp { preparable :: m -> Bool, readout :: m -> [Double]
, units :: [Unit], resolution, window :: Double }

data BareObject m = BareObject (Kinematic m) (Generator m) (Flow m)
data TheoryObject m = TheoryObject (BareObject m) (Interp m)
forget :: TheoryObject m -> BareObject m -- the operation U
forget (TheoryObject b _) = b

-- Cartesian monoidal product: block-diagonal form, summed generator,
-- paired flow. 'split' evaluates omega_1 (+) omega_2 in the splitting.
tensor :: BareObject a -> BareObject b -> BareObject (a, b)
tensor (BareObject ka (Generator ha) fa) (BareObject kb (Generator hb) fb) =
```

```

BareObject (Kinematic (dimHalf ka + dimHalf kb) split)
  (Generator (\(x,y) -> ha x + hb y))
  (\t (x,y) -> (,) <$> fa t x <*> fb t y)
where split (x,y) (v,w) = let n = 2 * dimHalf ka
  (v1,v2) = splitAt n v; (w1,w2) = splitAt n w
  in symplForm ka x (v1,w1) + symplForm kb y (v2,w2)

-- The non-interaction result as a sampled PREDICATE, not a theorem: a
-- coupling is non-interacting exactly when its differential vanishes.
-- This names the criterion; it does not prove the equivalence.
newtype Coupling m = Coupling (Observable m)
isNonInteracting :: (Observable m -> m -> [Double]) -> Coupling m -> [m] -> Bool
isNonInteracting grad (Coupling v) zs = all (all ((< 1e-12) . abs) . grad v) zs

-- Morphisms: the type records the data, it does not enforce the
-- immersion lemma, which is proved in the text and not here.
data Morphism a b = Morphism { mapPoints :: a -> b, shiftConst :: Double }

```

B Source verification record

Author list, title, year, and venue were matched against publisher or repository records for every cited item before citation. Confirmed without material residual uncertainty: [3] (LNP 813, 2011, pp. 95–174), [4] (Trans. AMS **50**, 1941, 71–128), [5] (TAC **30**, 2015, no. 33, 1096–1120), [7] (Invent. Math. **82**, 1985, 307–347, DOI 10.1007/BF01388806), [8] (Springer GTM 5, 2nd ed., 1998), [11] (Living Rev. Relativ. **17**, 2014, 4; arXiv:1403.7377, where the figure 42.98''/century appears). Confirmed with a stated residual: [1] (Benjamin/Cummings, 2nd ed., 1978, ISBN 978-0-8053-0102-1 — chapter pointers are to that edition and no page numbers are asserted); [2] (Springer GTM 60, 2nd ed., 1989, trans. Vogtmann & Weinstein — the pointers §43 and Ch. 10 were not independently re-paginated); [9] (Springer TAM 17, 2nd ed., 1999, ISBN 978-0-387-98643-2 — chapter pointers to that edition); [6] (arXiv:math-ph/9809011, 1998 — the preprint is cited because the published-volume pagination was not confirmed); [10] (Port. Math. **67**, 2010, start page 261 confirmed — the end page 278 is inferred and not confirmed).

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